Q1 (2) - Single Correct

We have, $f(x)+2f\left(\frac{2002}{x}\right)=3x$ ...(i)

Replacing x by $\frac{2002}{x}$ in Eq. (i), we get

$f\left(\frac{2002}{x}\right)+2f(x)=\frac{6006}{x}$ ...(ii)

On multiplying by 2 in Eq. (ii) and then subtracting from Eq. (i), we get

$$\begin{aligned} -3f(x) &= 3x-\frac{2\times 6006}{x} \\ -f(x) &= x-\frac{4004}{x} \\ f(x) &= \frac{4004}{x}-x \\ \therefore f(2) &= \frac{4004}{2}-2=2000 \end{aligned}$$

Q2 (3) - Single Correct

We know that, AM $\geq$ GM

then $\frac{a^2+b^2}{2}\geq\sqrt{a^2b^2}$

$\because a^2+b^2\geq 2ab$

Similarly, $b^2+c^2\geq 2bc$

and $c^2+a^2\geq 2ca$

On adding Eq. (i), (ii) and (iii), we get

$2\left(a^2+b^2+c^2\right)\geq 2(ab+bc+ca)$

$\therefore \quad ab+bc+ca\leq 1$

also $a^2+b^2-c^2<2ab$

$b^2+c^2-a^2<2bc$

$c^2+a^2-b^2<2ac$

On adding $a^2+b^2+c^2<2(ab+bc+ca)$

$\frac{1}{2}<ab+bc+ca$

$\therefore ab+bc+ca\in\left(\frac{1}{2},1\right]$

Q3 (2) - Single Correct

We have,

$f(x)=\sqrt{\frac{8}{1-x}+\frac{8}{1+x}}=\sqrt{\frac{16}{1-x^2}}=\frac{4}{\sqrt{1-x^2}}$

$\therefore \quad f(x)=\dfrac{4}{\sqrt{1-x^2}}$

$f(\sin x)=\dfrac{4}{|\cos x|}$ and $f(\cos x)=\dfrac{4}{|\sin x|}$

Now, we have

$g(x)=\frac{4}{f(\sin x)}+\frac{4}{f(\cos x)}=\frac{4|\cos x|}{4}+\frac{4|\sin x|}{4}$

$\therefore g(x)=|\sin x|+|\cos x|$

which is even function and both are complementary to each other.

Hence, period of $g(x)=\frac{\pi}{2}$.

Q4 (1) - Single Correct

Put $x=30, y=\frac{40}{30}$ then

$\mathrm{f}\left(30\times\frac{40}{30}\right)=\frac{\mathrm{f}(30)}{40}\times 30$

Hence,

$f(40)=15$

Q5 (3) - Single Correct

We have, $f(x)=\sin^2 x+\cos^4 x+2$

Its period is T_1.

Now, $f(x)=\frac{1-\cos 2x}{2}+\left(\frac{1+\cos 2x}{2}\right)^2+2$

$=\frac{1-\cos 2x}{2}+\frac{1}{4}\left(1+\cos^2 2x+2\cos 2x\right)+2$

$=\frac{1}{4}\left[2-2\cos 2x+1+\cos^2 2x+2\cos 2x+8\right]$

$=\frac{1}{4}\left[11+\cos^2 2x\right]=\frac{1}{4}\left[11+\frac{1+\cos 4x}{2}\right]$

$=\frac{1}{8}[22+\cos 4x]$

$\therefore \quad T_1=\frac{2\pi}{4}=\frac{\pi}{2}$

We have,

$g(x)=\cos(\cos x)+\cos(\sin x)$

Its period is T_2,

$\because g\left(x+T_2\right)=g(x)$

$\cos[\cos(x+T_2)]+\cos[\sin(x+T_2)]=\cos(\cos x)+\cos(\sin x)$

Put $x=0$

$$\cos(\cos T_2)+\cos(\sin T_2)=\cos 1+\cos 0$$

$\therefore \cos(\cos T_2)+\cos(\sin T_2)=\cos\left(\cos\frac{\pi}{2}\right)+\cos\left(\sin\frac{\pi}{2}\right)$

On comparing both sides, we get

$\therefore T_2=\frac{\pi}{2}$

We see that

$T_1=T_2$

Q6 (2) - Single Correct

$|x-p|=x-p$, since $x\geq p$

$$\begin{aligned} |x-15| &= 15-x && [\because x\leq 15] \\ |x-(p+15)| &= (p+15)-x && [\because 15+p>x] \end{aligned}$$

$\therefore$ Expression reduces to

$$\begin{aligned} E &= (x-p)+(15-x)+(p+15-x) \\ &= 30-x \end{aligned}$$

$\therefore E_{\min}$ Occurs, when $x=15$

$\Rightarrow E_{\min}=15$

Q7 (4) - Single Correct

We have,

$$\begin{aligned} g\left(-1,-\frac{3}{2}\right) &= -1-\left(-\frac{3}{2}\right)=\frac{1}{2} \\ g(-4,-1.75) &= -1.75-(-4) \\ &= 2.25 \end{aligned}$$

$$\text{Now, } f\left(g\left(-1,-\frac{3}{2}\right), g(-4,-1.75)\right)=(2.25)^{\frac{1}{2}}=1.5$$

Q8 (4) - Single Correct

We have to find solution set of

$$(f(x))^2 \geq f(x) + 6$$
$$(f(x))^2 - f(x) - 6 \geq 0$$
$$(f(x) - 3)(f(x) + 2) \geq 0$$
$$\Rightarrow \quad f(x) \leq -2 \text{ or } f(x) \geq 3$$

We have, $f(x) < 3$ solution set is $0 < x < \infty$.

$\therefore$ Solution set of $f(x) \geq 3$ is $-\infty < x \leq 0$ and $f(x) > -2$ solution set $-\infty < x < 5$

$\therefore f(x) \leq -2$

It solution set $5 \leq x < \infty$

Hence, complete solution set is

$(-\infty, 0] \cup [5, \infty)$

Q9 (4) - Single Correct

$|g(x)| - 1 \mid = \dfrac{1}{2}$

$|g(x)| - 1 = \pm\dfrac{1}{2}$

$|g(x)| = \dfrac{3}{2}$ and $|g(x)| = \dfrac{1}{2}$

$g(x) = \dfrac{\pm 3}{2}$ and $g(x) = \dfrac{\pm 1}{2}$

Hence $|g(\mathrm{x})| = \frac{1}{2}$ and $|g(\mathrm{x})| = \frac{3}{2}$.

Now consider the graph of $y = |g(x)|$.

If we draw a straight line parallel to x axis passing through $y = \frac{1}{2}$,

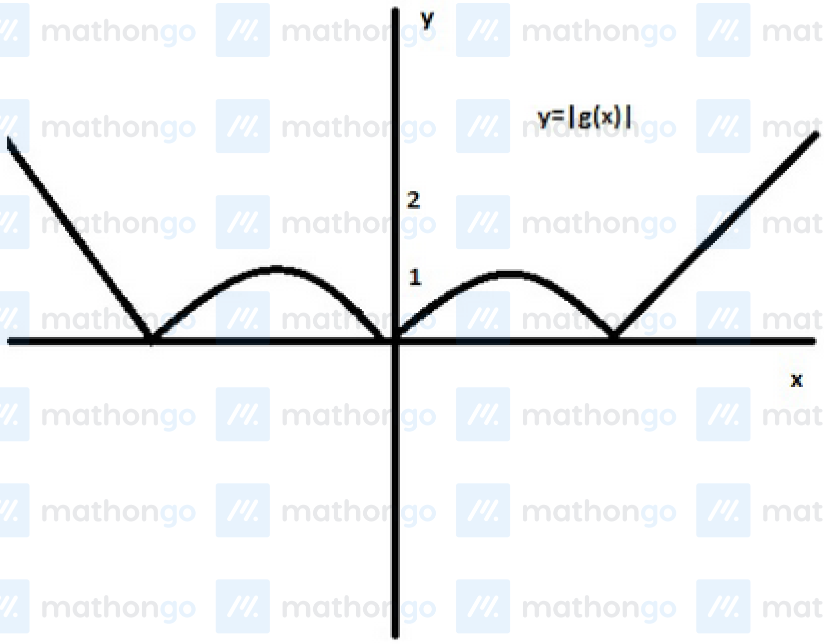
we get 6 points where the straight line cuts the the graph of $y = |g(x)|$.

Hence we get 6 solutions to the equation of $|g(x)| = 0.5$.

Now if we draw the a straight line passing through $y = 1.5$ and parallel to x axis,

then the straight line cuts the graph of $y = |g(x)|$ at exactly 2 points.

Hence $|g(x)|=1.5$ has 2 solutions.



Q10 (2) - Single Correct

We have, $f(x)=x^2+bx+c$

and $f(2+t)=f(2-t)$

$(2+t)^2+b(2+t)+c=(2-t)^2+b(2-t)+C$

$\Rightarrow \quad b=-4$

$\therefore \quad f(x)=x^2-4x+c$

Now, $\quad f(1)=c-3$

$f(2)=c-4$

$f(4)=c$

We see that, $f(2)<f(1)<f(4)$

Q11 (4) - Single Correct

We have, $[x]\{x\}=1$

$\{x\}=\frac{1}{[x]} \quad [0 \leq \{x\}<1]$

$\therefore \quad 0 \leq \frac{1}{[x]}<1$

$\therefore \quad 1<[x]<\infty \Rightarrow 2 \leq x<\infty$

But $\quad \{x\}=\frac{1}{[x]}, \{x\}=\frac{1}{m}$

$\therefore$ We see that, if $[x]=m$ Then, $\{x\}=\frac{1}{m}$

$\therefore x=[x]+\{x\}=m+\frac{1}{m}, m \in N$

But $m \neq 1$ i.e. $\left\{x=m+\frac{1}{m}: M \in N-\{1\}\right\}$

Q12 (2) - Single Correct

We have, $fog(x)=\operatorname{gof}(x)$

$\therefore \quad \frac{g(x)}{1+g(x)}=\frac{r f(x)}{1-f(x)}$

$\Rightarrow \quad \frac{\frac{r x}{1-x}}{1+\frac{r x}{1-x}}=\frac{\frac{r x}{1+x}}{1-\frac{x}{1+x}}$

$\Rightarrow \quad \frac{r x}{1-x+r x}=\frac{r x}{1}$

$\Rightarrow r x-r x(1-x+r x)=0$

$r x(1-1+x-r x)=0$

$r[x(1-r)]=0$

$\therefore r(r-1)=0$

$r=0$ and 1

Number of elements in set $S=2$.

Q13 (3) - Single Correct

In the functional relation replace x with $\mathrm{x}+1$.

We have,

$f(1+x)=f(1-x)$

This shows that the function is symmetric about $x=1$.

There is one and only one double root. If the double root exists at any value x_0 other than at $\mathrm{x}=1$, then a double root will also exist at a value of $2-\mathrm{x}_0$.

Hence, the double root exists at $\mathrm{x}=1$

Say two other roots are α and β

$f(\alpha)=f(2-\alpha)=0$

$\therefore 2-\alpha$ is also a root.

And similarly, $2-\beta$ is also a root.

$\therefore$ the roots are $1, 1, \alpha, \beta, 2-\alpha, 2-\beta$

Hence, sum of the roots is 6

Q14 (1) - Single Correct

Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots. + a_{n-1}x + a_n$

$f'(x) = a_0nx^{n-1} + a_1(n-1)x^{n-2} + \ldots.. + a_{n-1}$

$f''(x) = a_0n(n-1)x^{n-2} + a_1(n-1)(n-2)x^{n-3} + \ldots. + a_{n-2}$

$f(3x) = 3^na_0x^n + 3^{n-1}a_1x^{n-1} + 3^{n-2}a_2x^{n-2} + \ldots. + 3a_{n-1}x + a_n$

$f'(x)f''(x) = \left[a_0nx^{n-1} + a_1(n-1)x^{x-2} + \ldots. + a_{n-1}\right]\left[a_0n(n-1)x^{n-2} + a_1(n-1)(n-\right.$

$\left.2)\ x^{n-3} + \ldots. + a_{n-2}\ \right]$

Comparing highest powers of x,

$3^na_0x^n = a_o^2(n-1)x^{n-1+n-2} = a_0^2n^2(n-1)x^{2n-3}$

$\therefore 2n - 3 = n$

$\Rightarrow$ n $= 3$

and $3^na_o = a_o^2n^2(n-1)$

$3^3 = a_09(3-1)$

$a_0 = \frac{27}{18} = \frac{3}{2}$

$\therefore f(x) = \frac{3}{2}x^3 + a_1x^2 + a_2x + a_3$

$f(3x) = \frac{81}{2}x^3 + 9a_1x^2 + 3a_2x + a_3$

$f'(x) = \frac{9}{2}x^2 + 2a_1x + a_2$

$f''(x) = 9x + 2a_1$

$\frac{81}{2}x^3 + 9a_1x^2 + 3a_2x + a_3 = \left(\left(\frac{9}{2}x^2 + 2a_1x + a_2\right)(9x + 2a_1)\right)$

$\frac{81}{2}x^3 + 9a_1x^2 + 3a_2x + a_3 = \frac{81}{2}x^3 + [9a_1 + 18a_1]x^2 + \left[4a_1^2 + 9a_2\right]x + 2a_1a_2$

Comparing coefficients,

$\mathrm{aa_1} = 27\mathrm{a_1}$

$\Rightarrow \mathrm{a_1} = 0$

$3a_2 = 4a_1^2 + 9a_2 = 9a_2$

$\Rightarrow \mathrm{a}_2 = 0$

$a_3 = 2a_1a_2 \Rightarrow a_3 = 0$

$\therefore f(x) = \frac{3}{2}x^3$

$f'(x) = \frac{9}{2}x^2$

$f''(x) = 9x$

$f''(2) - f'(2) - 18 - 18 = 0$

Q15 (4) - Single Correct

Since, f is many one. Let $f(\alpha) = f(\beta)$, where $\alpha \neq \beta$.

Now, $f(\alpha + y) = g[f(\alpha), y] = g[f(\beta), y] = f(\beta + y)$

$\therefore g$ is periodic.

Also for $y = t - \alpha$

$\Rightarrow f(t) = f(t + \beta - \alpha), \forall t \in R$

$\therefore f$ is periodic.

Q16 (2) - Multiple Correct

Given, $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x}$ ...(i)

Replacing x by $\frac{1}{1-x}$ and by $\left(1 - \frac{1}{x}\right)$ in Eq. (i) one by one and on solving, we get

$$f(x) = \frac{x+1}{x-1}$$

$\therefore f(2) + f(3) = 3 + 2 = 5$

also, $f(x) = \frac{x+1}{x-1}$ crosses Y -axis at $x = 1$

Q17 (1, 2) - Multiple Correct

We have,

$$g(x) = \sin\left(\sin^{-1}\sqrt{\{x\}}\right) + \cos\left(\sin^{-1}\sqrt{\{x\}}\right) - 1$$
$$= \sqrt{\{x\}} + \sqrt{1 - \{x\}} - 1$$

Case I

If $x \in I$

$\Rightarrow g(x) = 0 + 1 - 1 = 0$

$\Rightarrow g(-x) = g(x)$

Case II

If $x \notin I$,

$$\Rightarrow \quad g(-x)=\sqrt{\{-x\}+\sqrt{1-\{-x\}}-1}$$
$$=\sqrt{1-\{x\}+\sqrt{1-(1-\{x\})}-1}$$
$$=\sqrt{\{x\}}+\sqrt{1-\{x\}}-1=g(x)$$

$\therefore g$ is even function and period is 1.

Q18 (3) - Multiple Correct

We have, $f(x)=\sin\frac{\pi}{4}[x]+\cos\frac{\pi x}{2}+\cos\frac{\pi}{3}[x]$

Let $T_1=$ Period of $\sin\frac{\pi}{4}[x]$

$\Rightarrow \quad \sin\frac{\pi}{4}[x+T_1]=\sin\frac{\pi}{4}[x]$

$\Rightarrow \quad \frac{\pi}{4}[x+T_1]=2n\pi+\frac{\pi}{4}[x]$

$\Rightarrow [x+T_1]= \quad 8n+[x]$

$\therefore T_1=8n$

Its minimum value is 8.

Hence, $T_1=8$

$T_2=$ Period of $\cos\frac{\pi x}{2}=\frac{2\pi}{\frac{\pi}{2}}=4$

$T_3=$ Period of $\cos\frac{\pi}{3}[x]=6$

$\therefore$ Period of $[x]=\angle CM$ of T_1, T_2 and $T_3=24$

Q19 (1, 2, 3) - Multiple Correct

We have, $f(x)=[x]+\left[x+\frac{1}{3}\right]+\left[x+\frac{2}{3}\right]-3x+15$

$=-x+[x]-\left(x+\frac{1}{3}\right)+\left[x+\frac{1}{3}\right]-\left(x+\frac{2}{3}\right)+\left[x+\frac{2}{3}\right]+15+\frac{1}{3}+\frac{2}{3}$

$\Rightarrow \quad f(x)=-\{x\}-\left\{x+\frac{1}{3}\right\}-\left\{x+\frac{2}{3}\right\}+16$

$\Rightarrow f\left(x+\frac{1}{3}\right)=-\left\{x+\frac{1}{3}\right\}-\left\{x+\frac{2}{3}\right\}-\{x+1\}+16$

$=-\left\{x+\frac{1}{3}\right\}-\left\{x+\frac{2}{3}\right\}-\{x\}+16$ as $\{x+1\}=\{x\}$

$\Rightarrow f\left(x+\frac{1}{3}\right)=f(x)$

$\therefore$ Period of $f(x)$ is $\left(\frac{1}{3}\right)$

Q20 (1) - Multiple Correct

We have,

$$f(x) = (x+a) - [x+b] + \sin \pi x + \cos 2\pi x$$
$$+ \sin 3\pi x + \cos 4\pi x + \cdots + \sin(2n-1)\pi x + \cos 2n\pi x$$

of $\quad x + a - [x+b] + b - b$

To period of

$\Rightarrow x + b - [x+b] + a - b$

$\Rightarrow \{x+b\} + a - b$

Hence, we see that its period is 1.

Now, period of $\sin \pi x$ is $\frac{2\pi}{\pi} = 2$ Period of $\cos 2\pi x$ is $\frac{2\pi}{2\pi} = 1$

Similarly, period of $\cos 2n\pi x = \frac{2\pi}{2n\pi} = \frac{1}{n}$.

$\therefore$ Period of $f(x)$ is LCM of all above period, which is 2.

Q21 (3) - Multiple Correct

We have, $f(x) = \sin\left(\frac{\pi}{2}[x]\right)$

To check $f(x)$ is even or odd.

$f(-x) = \sin\left(\frac{\pi}{2}[-x]\right)$

Clearly, we cannot say anything because x is not given integer or not integer.

Hence, $f(x)$ is not even or odd.

To check periodicity

$$f(x+T) = f(x)$$
$$f(x+T) = \sin\left(\frac{\pi}{2}[x+T]\right) = \sin\left(\frac{\pi}{2}[x]\right)$$

$\Rightarrow \quad \frac{\pi}{2}[x+T] = 2n\pi + \frac{\pi}{2}[x]$

$\Rightarrow \quad [x+T] = 4n + [x]$

$\Rightarrow \quad T + [x] = 4n + [x]$

$\therefore T = 4n$

Hence, $f(x)$ is periodic with period 4.

Q22 (1, 2, 3) - Multiple Correct

(1)$f(x)=\cos\left(\cos^{-1}x\right)=x; -1\leq x\leq 1$, then

$f(-x)=\cos\cos^{-1}(-x)=-x$

$\therefore$ It is odd function.

Hence, it is incorrect.

(2) We have,

$f(x)=\cos(\sin x)+\cos(\cos x)$

Let it period be T.

$$f(x+T)=f(x)$$
$$\cos(\sin(x+T))+\cos(\cos(x+T))=\cos(\sin x)+\cos(\cos x)$$

Put $x=0$, we get

$$\cos\sin T+\cos\cos T=\cos 0+\cos 1$$
$$\text{or } \cos\sin T+\cos\cos T=\cos\left(\cos\left(\frac{\pi}{2}\right)\right)+\cos\left(\sin\left(\frac{\pi}{2}\right)\right)$$

By comparing, we get $T=\frac{\pi}{2}$

Hence, its period is $\frac{\pi}{2}$

It is incorrect.

(3) We have, $f(x)=\cos(3\sin x)$

$-3\leq 3\sin x\leq 3$

range of $\cos(3\sin x)=[\cos(-3),\cos(3)]$

Q23 (2, 3) - Multiple Correct

We have, $f(x)=\frac{\sin\pi[x]}{\{x\}}$

Let T_1 be the period of $\sin\pi[x]$.

Then, $\sin\pi\left[T_1+x\right]=\sin\pi[x]$

$\Rightarrow \pi\left[T_1+x\right]=2n\pi+\pi[x]$

$\Rightarrow T_1+[x]=2n+[x]$

$\therefore T_1=2n$

and minimum value is 2.

$T_1=2$

$T_2=$ Period of $\{x\}$ is 1.

Hence, period of $f(x)=LCM$ of $(T_1$ and $T_2)$

$= 2$

To find range of $f(x) = \sin \pi[x]$ is always 0.

Hence, range of $f(x) = 0$, which is a singleton set.

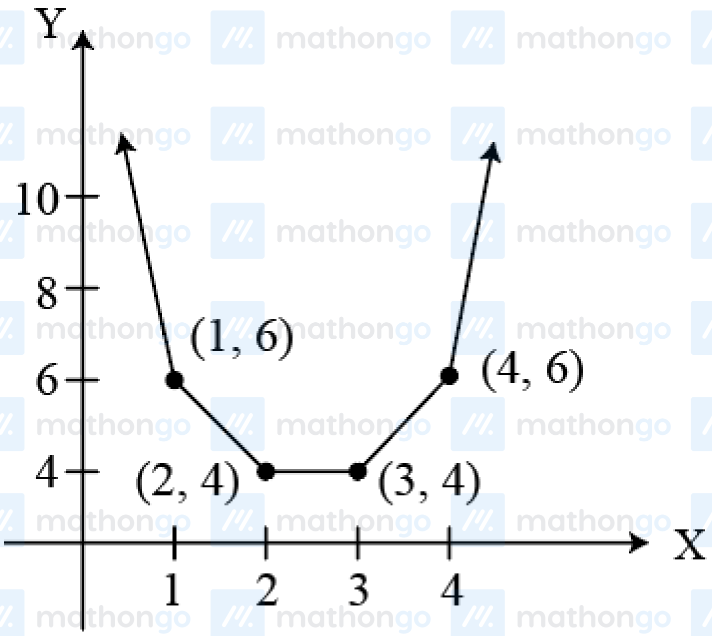
Since, $f(x)$ is always $0, \forall x \in R$

$\therefore f(x)$ is an even function.

Q24 (1, 2, 3) - Multiple Correct

We have, $f(x) = |x-1| + |x-2| + |x-3| + |x-4|$

$$f(x) = \begin{cases} -4x+10; & x < 1 \\ -2x+8; & 1 \leq x < 2 \\ 4; & 2 \leq x < 3 \\ 2x-2; & 3 \leq x < 4 \\ 4x-10; & 4 \leq x < \infty \end{cases}$$



Clearly, we see that $f(x)$ is least i.e. 4, which is not unique and its least value is 4

Q25 (1, 4) - Multiple Correct

We have, $f(x) = \sec^{-1}\left[1 + \cos^2 x\right]$

We know that domain of $\sec^{-1} x$ is $|x| \geq 1$

$\because \left|\left[1 + \cos^2 x\right]\right| \geq 1$

which is true for every $x \in R$ To find the range $0 \leq \cos^2 x \leq 1 \Rightarrow 1 \leq (1+\cos^2 x) \leq 2$

$\Rightarrow \quad [1+\cos^2 x] = 1$ and 2

$\therefore$ Range of $f(x)$ is $\{\sec^{-1} 1, \sec^{-1} 2\}$

Q26 (1) - Paragraph 1

We have, $f(x) = \frac{x+a}{bx^2+cx+2}$, where $a, b, c \in R, f(-1) = 0$ and

$y = 1$ is an asymptote of $y = f(x)$

$\therefore a = 1, b = 0, c = 1 \Rightarrow f(x) = \frac{x+1}{x+2}$

Q27 (2) - Paragraph 1

We have, $f^{-1}(x) = \frac{2x-1}{1-x}$

Now, $f^{-1}(x) : R - \{-1\} \to R$

and $y = \frac{2x-1}{1-x} \Rightarrow x = \frac{y+1}{y+2}; y \neq -2$

$\therefore$ Range of $f^{-1}(x)$ is $R - \{-2\} \subset$ Codomain

Thus, f is one-one and into.

Q28 (2) - Paragraph 2

We have, $f(x) = x^2 - 2x - 1$

$\therefore f'(x) = 2x - 2$

For injective, $f'(x) \geq 0$

or $f'(x) \leq 0, \quad \forall x \in (-\infty, a)$

For $f'(x) \leq 0 \Rightarrow \quad 2x - 2 \leq 0$

$x \leq 1$

$\therefore$ Largest value of a is 1.

For surjective,

$f(x) = x^2 - 2x - 1$

$f(x) = (x-1)^2 - 2$

$\Rightarrow f(x)$ is surjective, if $f(x) \in [-2, \infty)$

$\therefore a=1$ and $b=-2$

and $a+b=1-2=-1$

Q29 (3) - Paragraph 2

We have, $g(x)=f(x)+3x-1$

$\Rightarrow g(x)=x^2-2x-1+3x-1\left[\because f(x)=x^2-2x-1\right]$

$\Rightarrow g(x)=x^2+x-2$

and $y=g|x|$

$\Rightarrow y=|x|^2+|x|-2$

$\therefore$ The least value of y is -2 ($\because |x|\geq 0$)

Q30 (3) - Paragraph 3

Explanation (For Question 30, 31)

We have, $f_1(x)=e^{\tan\left\{\frac{x}{4}\right\}}+\cos\pi\left(\frac{1-2[x]}{2}\right)+\sin\left(\frac{\pi[x]}{2}\right)$

Let $\quad T_1=$ Period of $e^{\tan\left\{\frac{x}{4}\right\}}=\frac{1}{\frac{1}{4}}=4$

$T_2=$ Period of $\cos\pi\left(\frac{1-2[x]}{2}\right)$

or $\quad \cos\left(\frac{\pi}{2}-\pi[x]\right)=\sin\frac{\pi[x]}{2}$

$\Rightarrow \qquad \sin\frac{\pi}{2}[x+T_2]=\sin\frac{\pi}{2}[x]$

$\Rightarrow \qquad \frac{\pi}{2}[x+T_2]=2n\pi+\frac{\pi}{2}[x]$

$\Rightarrow \qquad [x+T_2]=4n+[x] \quad \Rightarrow \quad T_2=4n$

$\because \quad$ Its period $T_2=4$

So, period of $f_1(x)$ is $p=4$ We have, $y=\sqrt{2p+\frac{p}{2}[x]-[x]^2}$

We calculate $\quad p=4$

$\therefore y=\sqrt{8+2[x]-[x]^2}$

Domain $=8+2[x]-[x]^2\geq 0$

$\Rightarrow [x]^2-2[x]-8\leq 0$

$\Rightarrow([x]-4)([x]+2)\leq 0$

$\Rightarrow -2\leq[x]\leq 4$

$\Rightarrow -2\leq x<5$

$[n_2 \leq [x] \leq n_1 \Rightarrow n_2 \leq x < n_1 + 1, \text{ where } n, n_2 \in l]$

Since, we have domain of y is $[q, r)$

Hence, by comparing, we get $q = -2$ and $r = 5$

$p = 4$, which is a composite number.

Q31 (1) - Paragraph 3

The value of $r - q - 1$

Q32 (2) - Paragraph 4

Explanation (For Question 32, 33)

From the paragraph, the function is given by

$$f(x) = \begin{cases} \pi(2R - x)^2; & 0 \leq x \leq R \\ \pi R^2; & R \leq x \leq 3R \\ \pi\left(6Rx - x^2 - 8R^2\right); & 3R \leq x \leq 4R \end{cases}$$

We see that, $f(x)$ is not defined outside the interval $[0, 4R]$ $f(x)$ is decreases on $[0, R]$, so $f(x)$ takes values in $\left[\pi R^2, 4\pi R^2\right]$ for $x \in [0, R]$

We see that on the interval [3R, $4R$], $f(x)$ is decreases, so $f(x)$ takes values in $\left[0, \pi R^2\right]$. Hence, range of f is $\left[0, 4\pi R^2\right]$, since $f(x)$ is strictly decreasing function on $[0, R] \cup [3R, 4R]$, so it is one-one. On this interval, moreover,the function is constant on $[R, 3R]$, thus it cannot be one-one on $[R, 3R]$ or a set containing $[R, 3R]$.

Q33 (1) - Paragraph 4

Q34 (1) - Paragraph 5

Explanation (For Question 34, 35)

Here, $\{f(x)\}^2 \cdot f\left(\frac{1-x}{1+x}\right) = 64x$ ...(i)

Putting $\frac{1-x}{1+x} = y$ and $x = \frac{1-y}{1+y}$, we get

$$\left\{f\left(\frac{1-y}{1+y}\right)\right\}^2 \cdot f(y) = 64 \cdot \left(\frac{1-y}{1+y}\right)$$

$\Rightarrow \quad f(x) \cdot \left(f\left(\frac{1-x}{1+x}\right)\right)^2 = 64\left(\frac{1-x}{1+x}\right)$ ...(ii)

From Eqs. (i) and (ii), we get

$$\frac{\{f(x)\}^4\cdot\left\{f\left(\frac{1-x}{1+x}\right)\right\}^2}{f(x)\cdot\left\{f\left(\frac{1-x}{1+x}\right)\right\}^2}=\frac{(64x)^2}{64\left(\frac{1-x}{1+x}\right)}$$

$\Rightarrow \quad \{f(x)\}^3 = 64x^2 \cdot \left(\frac{1+x}{1-x}\right)$

$\Rightarrow \quad f(x) = 4x^{2/3} \cdot \left(\frac{1+x}{1-x}\right)^{1/3}$

Q35 (2) - Paragraph 5

Q36 (3) - Integer Type

$\because f(x) = [x] + \sum_{r=1}^{2015} \frac{\{x+r\}}{2015}$

$= [x] + \frac{\{x+1\}+\{x+2\}+\ldots+\{x+2015\}}{2015}$

$= [x] + \frac{2015\{x\}}{2015} = [x] + \{x\}$

$= x$

$\therefore \quad f(3) = 3$

Q37 (2) - Integer Type

We have,

$f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ ...(i)

Replacing x by $x+a$ in Eq. (i), we get

$f(x+a+a) = \frac{1}{2} + \sqrt{f(x+a) - (f(x+a))^2}$ ...(ii)

On putting the value of $f(x+a)$ from Eq. (i) in Eq. (ii), we get

$f(x+2a) = \frac{1}{2} + \sqrt{\frac{1}{2} + \sqrt{f(x) - \{f(x)\}^2} - \left\{\frac{1}{2} + \sqrt{f(x) - \{f(x)\}^2}\right\}^2}$

$\Rightarrow f(x+2a) = \frac{1}{2} + \sqrt{\frac{1}{2} + \sqrt{f(x) - \{f(x)\}^2} - \frac{1}{4} - f(x) + (f(x))^2 - \sqrt{f(x) - \{f(x)\}^2}}$

$\Rightarrow \quad f(x+2a) = \frac{1}{2} + \sqrt{\frac{1}{2} - \frac{1}{4} - f(x) + \{f(x)\}^2}$

$= \frac{1}{2} + \sqrt{\frac{1}{4} - f(x) + \{f(x)\}^2}$

$= \frac{1}{2} + \sqrt{\left(\frac{1}{2} - f(x)\right)^2}$

$= \frac{1}{2} + f(x) - \frac{1}{2}$

$\therefore f(x+2a)=f(x)$

$\therefore$ Its period is $2a$ but it is given λ a. Hence, $\lambda=2$.

Q38 (8) - Integer Type

We have,

$f(x-1)+f(x+3)=f(x+1)+f(x+5)$ ...(i)

Replacing x by $x+2$ in Eq. (i), we get

$f(x+1)+f(x+5)=f(x+3)+f(x+7)$ ...(ii)

On adding Eqs. (i) and (ii), we get

$f(x-1)=f(x+7)$ ...(iii)

Now, replacing x by $x+1$ in Eq. (iii), we get

$f(x)=f(x+8)$

Hence, period of $f(x)$ is 8.

Q39 (1) - Integer Type

Given, $\frac{(\sin x)^{2y}}{(\cos x)^{y^2/2}}+\frac{(\cos x)^{2y}}{(\sin x)^{y^2/2}}\geq 2\sqrt{\frac{(\sin x\cos x)^{2y}}{(\cos x\sin x)^{y^2/2}}}$ [AM $\geq$ GM]

$\geq 2(\sin x\cos x)^{y-\frac{y^2}{4}}$

$\Rightarrow\sin 2x\geq 2(\sin x\cos x)^{y-\frac{y^2}{4}}$

$\therefore y-\frac{y^2}{4}\geq 1$

$\Rightarrow y^2-4y+4\leq 0$

$\Rightarrow(y-2)^2\leq 0$

$\therefore y=2$ and $\sin x=\cos x$

$\Rightarrow x=\pi/4, y=2$

$\therefore$ Number of pairs (x,y) is 1

Q40 (1) - Integer Type

As, $f([i])=\frac{1}{1+[i]}$

$\Rightarrow\quad f(0)=1, f(1)=\frac{1}{2}, f(2)=\frac{1}{3}, f(3)=\frac{1}{4}, f(4)=\frac{1}{5}$

$g(x)=(1+x)f(x)-1$

is a polynomial of degree 5 with roots 0,1,2,3,4 $\Rightarrow \quad g(x)=Ax(x-1)(x-2)(x-3)(x-4)$

$\Rightarrow \quad A=\frac{1}{5!}$

$\Rightarrow \quad (1+x)f(x)-1=\frac{1}{5!}x(x-1)(x-2)(x-3)(x-4)$

$\therefore 7f(6)-1=\frac{6\times5\times4\times3\times2}{5!}$

$\Rightarrow 7f(6)-1=6$

$\Rightarrow f(6)=1$

Q41 (2) - Integer Type

Given, $f[xf(y)]=x^p y^4$

Put $x=\frac{1}{f(y)}$

$\therefore \quad f(1)=\left(\frac{1}{f(y)}\right)^p\cdot y^4=\frac{y^4}{[f(y)]^p}$

for $\quad y=1, f(1)=\frac{1}{[f(1)]^p}$

$\Rightarrow \quad f(1)=1$, So, $f(y)=y^{4/p}$

Hence, $f\left(xy^{4/p}\right)=x^p y^4$

Put $\dot{y}=z^{p/4}$

$\Rightarrow \quad f(xz)=x^\rho z^\rho$

$\Rightarrow f(x)=x^\rho$

$\therefore \frac{4}{p}=p$

$\Rightarrow p=2$

Q42 (4) - Integer Type

We have, $f(x)=2\cos^2 x+\sqrt{3}\sin 2x+1$

$\Rightarrow \quad f(x)=1+\cos 2x+\sqrt{3}\sin 2x+1$

$\Rightarrow \quad f(x)=2+2\left(\frac{1}{2}\cos 2x+\frac{\sqrt{3}}{2}\sin 2x\right)$

$\Rightarrow \quad f(x)=2+2\sin\left(2x+\frac{\pi}{6}\right)$

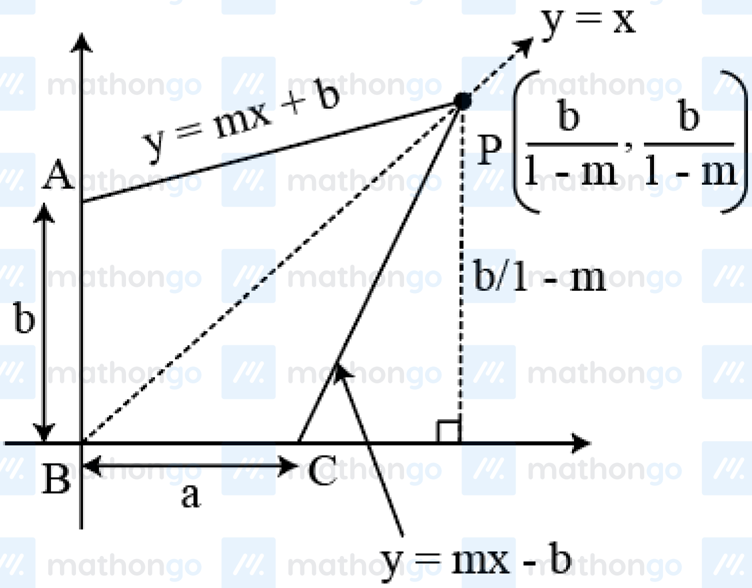
$\therefore$ Range of $f(x)=[0,4]$

Hence, $f^{-1}(x)$ exists, if $B\in[0,4]$, but $B\in[a,b]$.

Now, $(a+b)=(0+4)=4$

Q43 (3) - Integer Type

If $f(x)=mx+b$, then $f^{-1}(x)=\frac{x-b}{m}$ and their point of intersection can be found by setting $x=my+b$, since they intersect at $y=x$



Thus, $x=\frac{b}{1-m}$ and the point of intersection is $\left(\frac{b}{1-m}, \frac{b}{1-m}\right)$

Region R can be broken up into congruent Δ 's PAB and ΔPCB, which both have a base of b and a height of $\left(\frac{b}{1-m}\right)$

Area of $R=2\cdot\left(\frac{b}{2}\right)\cdot\left(\frac{b}{1-m}\right)=\frac{b^2}{1-m}=49$

For $m=\frac{9}{25} b^2=\frac{16}{25}\times 49 \Rightarrow b=\frac{28}{5}=\frac{\lambda}{\mu}$

$\therefore|\lambda-5\mu|=|28-25|=3$

Q44 (1) - Integer Type

We have,

$$\begin{aligned} & f(x)=x^2-4x+5 \\ \Rightarrow \quad & f(x)=(x-2)^2+1 \\ \therefore \quad & f(x)\in[1,\infty] \end{aligned}$$

But $f:[2,\infty)\rightarrow[1,\infty)$

Hence, the value of a is 1.

Q45 (6) - Integer Type

We have, $f(x)=\left(\frac{4a-7}{3}\right)x^3+(a-3)x^2+x+5$

$\therefore f'(x)=(4a-7)x^2+2(a-3)x+1$

For one-one function,

$f'(x)>0$ or $f'(x)<0$ for in domain

$\Rightarrow \quad f(x)>0$ iff $D<0$

$\Rightarrow \quad D=\{2(a-3)\}^2-4(4a-7)(1)<0$

$\Rightarrow \quad a^2-6a+9-4a+7<0$

$\Rightarrow \quad a^2-10a+16<0$

$\Rightarrow \left(a^2-8\right)(a-2)<0$

$\Rightarrow \quad a\in(2,8)$

But $a\in[\lambda,u]\Rightarrow\lambda=2,\mu=8$

$\therefore|\lambda-\mu|=|2-8|=6$

Q46 (6) - Integer Type

Number of different sets containing exactly 3 elements of

$B={}^5C_3=10$

Then, number of onto functions from A to the set containing

$$\begin{aligned}\text{3 elements }&=10\left[3^4-3\left\{(2)^4-2\right\}-3(1)^4\right]\\&=10[81-3(16)+3]\\&=10\times36=360\end{aligned}$$

$\therefore\frac{k}{60}=6$

Q47 (5) - Integer Type

Here, $f(x)-10x=(x-1)(x-2)(x-3)(x-\alpha)$

$\therefore \quad f(12)=120+(11\times10\times9)(12-\alpha)$

$f(-8)=-80+(-9)\times(-10)\times(-11)(-8-\alpha)$

$\Rightarrow f(12)+f(-8)= \quad 40+(9\times10\times11)\times20=19840=(3968)5$

$\therefore\lambda=5$

Q48 (0) - Integer Type

Given, $f(x)=\begin{cases}1+x^2, & x\leq 1\\ x+1, & 1<x\leq 2\end{cases}$

and $g(x)=1-x, -2\leq x\leq 1$

Now, $\text{fog}(x)=f\{g(x)\}=\begin{cases}1+\{g(x)\}^2, g(x)\leq 1\\ g(x)+1, 1<g(x)\leq 2\end{cases}$

$=\begin{cases}2-2x+x^2, & 0<x<1\\ 2-x, & -1\leq x<0\end{cases}$

On comparing, we get

$\therefore a+b+c=0+1+(-1)=0$

Q49 (6) - Integer Type

Given, $f(x)=\begin{cases}1+x, & 0\leq x\leq 2\\ 3-x, & 2<x\leq 3\end{cases}$

Now, (fof) $(x)=\begin{cases}1+f(x), & 0\leq f(x)\leq 2\\ 3-f(x), & 2<f(x)\leq 3\end{cases}$

$=\begin{cases}2+x_1 & -1\leq x\leq 1\\ 2-x, & 1<x\leq 2\\ 4-x, & 2<x\leq 4\end{cases}$

On comparing, we get

$a+b+c+d=-1+1+2+4=6$